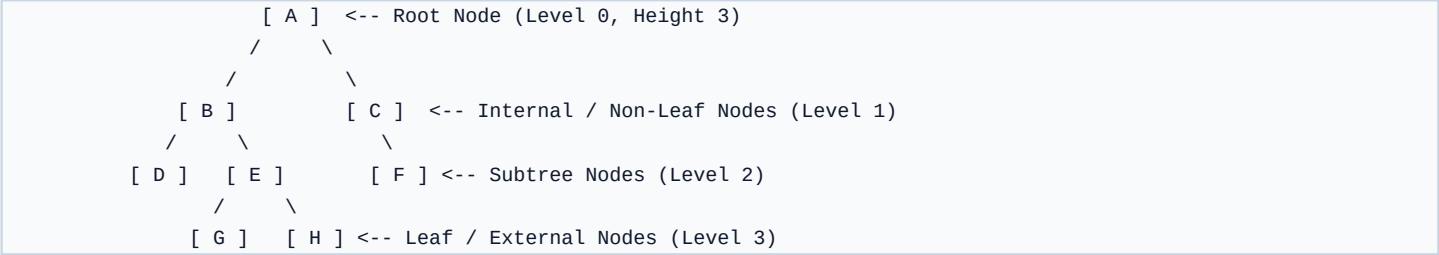


Tree Introduction & Terminologies -- Data Structures

Definition: A Tree is a non-linear, hierarchical data structure consisting of a collection of nodes connected by directed or undirected edges, such that there exists exactly one path between any two nodes and no cycles are formed.

1. Detailed Technical Explanation

Unlike linear data structures (Arrays, Linked Lists, Stacks, Queues) where elements are stored sequentially, trees organize data hierarchically.



Core Tree Terminologies:

1. Root: The topmost node in a tree with no parent (Node A).

2. Ed

2. Memory Keywords & Mathematical Relations

* Non-linear Hierarchy: Parent-child recursive relationship.

* No

3. Must-Write Points for Exams

* A tree with N nodes always has exactly N - 1 edges.

* Le

4. Quick Recall Flow

Hierarchical Structure -> Root Node -> Edges = N - 1 -> Degree (Child Count) -> Height/Depth -> Leaves (Degree 0)